

Chi-Yao (Trey) Huang

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[Personal Website](#)
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SUMMARY	Ph.D. researcher in representation learning, embodied AI, and world models. My research develops interpretable and controllable latent-space learning for embodied AI, connecting spatial perception, semantic understanding, and reasoning through structured visual representations. Previously a Senior R&D Engineer at HTC VIVE, where I developed core VO/SLAM tracking systems deployed across major commercial VR/AR product lines.		
TECHNICAL SKILLS	Research: World models, representation learning, contrastive learning, generative models, foundation models Computer Vision & Robotics: SLAM, visual odometry, 3D reconstruction, sensor fusion, path planning, aerial robotics, humanoid robotics Programming: Python, C/C++, JavaScript, MATLAB Tools & Frameworks: PyTorch, OpenCV, Git, ROS, Ceres, Eigen, Docker, OpenGL, SIMD optimization, CEVA/Hexagon DSP		
EDUCATION	Ph.D. in Computer Science Arizona State University, Tempe, AZ – Advisor: Prof. Yezhou Yang	2023 – PRESENT	
	M.S. in Robotics and Autonomous Systems Arizona State University, Tempe, AZ	2021 – 2023	
	M.S. in Mechanical Engineering National Taiwan University, Taipei, Taiwan	2015 – 2017	
	B.S. in Mechanical Engineering National Sun Yat-Sen University, Kaohsiung, Taiwan	2010 – 2014	
SELECTED PUBLICATIONS	Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning Chi-Yao Huang et al. <i>ICLR 2026</i> . Proposed an orthogonal latent-space construction framework that decomposes task-specific factors into structured subspaces to reduce representation collapse. (Web Paper Code Talk) VOCAL: Visual Odometry via ContrAstive Learning Chi-Yao Huang et al. <i>WACV 2026</i> . Proposed contrastive representation learning that organizes latent representations according to camera motion, producing interpretable motion-aware structure for visual odometry. (Web Paper Talk Demo)		
ONGOING RESEARCH	Domain World Models: Orthogonal Latent Dynamics and Heuristic Planning Developing an action-grounded latent-space framework for reward-free world models that organizes action-induced transitions along learned orthogonal directions, enabling interpretable and compositional latent dynamics and heuristic initialization for goal-directed planning. Object-Centric Learning with Contrastive Slot Attention Developing an object-centric representation-learning framework that integrates slot attention with contrastive learning to improve object discovery, slot consistency, and compositional visual reasoning.		
INDUSTRY EXPERIENCE	Senior R&D Engineer, Team Lambda <i>HTC VIVE, Advanced and Creative Team, New Taipei, Taiwan</i>	2017 – 2021	
	- Developed core VIO/SLAM tracking systems deployed across VIVE COSMOS, VIVE FOCUS 3, VIVE XR Elite, and VIVE FLOW. VIVE COSMOS: Architected a multi-camera tracking system for consumer VR, achieving sub-millimeter tracking precision in controlled evaluation; developed a high-performance “mothership” SLAM system. (Product Demo) VIVE FOCUS 3: Led prototyping for a flagship MR system and optimized a tightly coupled visual-inertial SLAM pipeline on resource-constrained embedded platforms, achieving a 4× runtime speed-up. (Product)		

- **VIVE XR Elite:** Engineered low-latency VIO and obstacle-avoidance systems for mixed-reality interaction; designed core data structures for real-time spatial computing. ([Product](#))
- **VIVE FLOW:** Designed inside-out tracking for limited-FoV sensors and developed a motion-tracking prototype integrating real-time hand tracking. ([Product](#))
- **Spatial Mapping:** Built voxel-based obstacle mapping and semantic mapping modules for VR safety and real-world interaction.

SELECTED PROJECTS

Toyota Mothership – Toyota Research Institute of North America (*Patent Pending*): Developed a UAV pose-estimation framework using infrared imagery and IMU sensor fusion for high-altitude drone perception under challenging lighting and weather conditions. The underlying technologies are the subject of patent-pending applications. ([Demo](#))

3D Landmark Reconstruction for Robot Localization and Mapping: Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for robot localization. ([Demo](#))

3D Reconstruction and Path Planning with Signed Distance Functions: Fused dense image information with sparse feature points for efficient SLAM; implemented voxel hashing for dense mapping and integrated reconstruction with robot path planning.

Humanoid Robot Control: Developed bipedal walking patterns on non-planar surfaces by optimizing zero-moment-point trajectories and implemented localized A* planning for obstacle navigation. ([Demo](#))

TEACHING & RESEARCH MENTORING

Graduate Research Mentoring

Mentored graduate students throughout their thesis research, providing guidance on problem formulation, literature review, experimental design, implementation, evaluation, and technical writing.

- **Zeel Bhatt (*OpalAI*)**
Thesis on contrastive representation learning for visual odometry and robotics.
- **Khoa (Albert) Vo (*Incoming Ph.D. student, Purdue University*)**
Thesis on multimodal reasoning and representation learning for embodied AI.
- **Shlok Mohanty (*Graduate Student*)**
Thesis on representation learning from imbalanced robotic task data.
- **Hoang Quy Nguyen (*Graduate Student*)**
Thesis on object-centric representation learning for structured visual understanding and downstream reasoning.

Teaching Assistant, CSE 571: Artificial Intelligence

Arizona State University

Supported course instruction through office hours, assignment grading, technical guidance, and individualized student support.

HONORS, AWARDS & GRANTS

Research Grant, Toyota Research Institute of North America

2023 – Present

Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University

Student with Distinction, Arizona State University

LANGUAGES

English (Fluent), Mandarin Chinese (Native)